Supply Chain Analysis:

Cass Beer Brewing Process

(Term Project Submission - Milestone 1)

Team4. 나태양, 유재호, 윤채리, 이대희, 최지안

1. Target Supply Chain

The selected supply chain is the **beer brewing process of Cass beer**.

This supply chain is characterized by:

- Long lead time (especially fermentation stage)
- High variability in raw material prices (e.g., barley, hops)
- Demand uncertainty influenced by seasonality and market trends

2. Product & Key Stakeholders

- **Product:** Beer
- **Stakeholders:**
  - Raw material suppliers (barley, hops providers)
  - Manufacturer (brewery company)
  - Distributors (logistics & wholesale)
  - Retailers (restaurants, convenience stores, bars)
  - Consumers

3. Scope Definition

- **Internal Scope:**
  - Raw material procurement
  - Five-stage brewing process
  - Inventory management
- **External Scope:**
  - Supply and distribution network
  - Regulatory environment (e.g., liquor tax laws)
  - Competitors' marketing strategies

4. Supply Chain Overview

Material Flow

Raw materials (barley, hops) → Brewing process (Work-in-Process) → Finished goods (beer) → Distribution network

Brewing Process (5 Stages)

1. **Mashing:** Extract sugars by mixing malt with water
2. **Boiling:** Add hops and boil to control flavor and bitterness
3. **Cooling:** Rapidly reduce temperature before yeast addition
4. **Fermentation:** Yeast converts sugar into alcohol
    - *This stage has the longest lead time and is a potential bottleneck*
5. **Packaging:** Carbonation, bottling/canning, and sealing

Information Flow

Customer demand data ↔ Demand forecasting ↔ Production scheduling ↔ Supplier ordering

Financial Flow

Revenue from beer sales ↔ Costs of raw materials and facility operations

5. Risk Analysis

- **Process Risk:**
  Production errors may lead to large-scale product disposal
- **Supply Risk:**
  Fluctuation in raw material prices
- **Demand Risk:**
  Uncertain and seasonal demand patterns

6. AI Opportunities

- **Quality Control:**
  AI-based monitoring to detect anomalies and prevent production errors
- **Procurement Optimization:**
  AI-driven prediction of raw material prices for strategic sourcing
- **Demand Forecasting:**
  Machine learning models to improve demand prediction accuracy